

Lesson 4: Direct Hit

GDevelop

VIDEO GAME CREATION

This week: detecting when things touch. Your job is to add one enemy to the screen and make your bullet destroy it on contact - the first piece of your capstone's actual combat loop.

Open your own [YourName]Game project from Lesson 3 - don't start a new one. Everything this week builds on the player and bullet you already have.

Tip: this lesson's Assets folder has a ready-to-use placeholder enemy image (enemy_invader_red.png) and a new sound effect (sfx_hit.wav) - you don't need to find or make your own art today.

Setup

1. Open your [YourName]Game project from Lesson 3.
2. Add a new Sprite object, name it Enemy, and assign enemy_invader_red.png (import it from this lesson's Assets folder) as its image.
3. Place one instance of Enemy on the scene near the top of the play area.

Level 1 - Minimal (MVP)

A single enemy sits on screen. When your bullet touches it, both the bullet and the enemy disappear. This is a complete finish for today - reaching this is enough to move on to next week.

4. Add a new event: Condition "Bullet is in collision with Enemy" -> Actions "Delete Bullet" and "Delete Enemy".
5. Click the green Preview button and shoot the Enemy to test - both the bullet and the Enemy should disappear on contact.

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Condition: Bullet is in collision with Enemy
Action: Delete Bullet
Action: Delete Enemy
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Level 2 - Complete (Alpha/Beta)

The enemy is destroyed cleanly and exactly once, even if you fire several bullets in a row - no errors, no lingering references, and a bullet that misses still behaves exactly like it did last week.

- Fire several bullets in a row, including some that miss - each hit deletes exactly one Bullet instance and the Enemy, and any bullet that misses keeps flying and is destroyed off-screen exactly like it did in Lesson 3 (the Destroy outside screen behavior).

- GDevelop's collision condition and Delete actions already operate per pair of overlapping instances each frame, so even if two bullets somehow overlap the Enemy on the same frame, the event simply runs twice that frame - the Enemy and both bullets are still deleted cleanly either way.

Level 3 - Refined (Ready to Publish) - pick any that interest you

- Sound: add a "Play a sound" action (using the imported sfx_hit.wav) to the same event as the collision condition.
- Visual: give Enemy its own "Tint color" (distinct from the bullet's yellow), applied once at the start of the scene, so it clearly reads as a threat rather than another shot.
- Structure: group this lesson's event (the collision condition and its actions) into its own labeled event group (e.g. "Combat"), separate from the "Shooting" group from Lesson 3.

If Something's Wrong

- Nothing happens on contact: confirm the collision condition is comparing Bullet against Enemy (not Player against Enemy), and that both objects are Sprite objects with an actual image assigned.
- Enemy disappears but a bullet keeps flying: confirm "Delete Bullet" is in the same event as the collision condition, not a separate always-running event.
- Nothing on screen at all: confirm at least one Enemy instance was actually placed on the scene, not just created as an object type - an object with zero instances can never collide with anything.

Want More?

Student Resource List > GDevelop section, or gdevelop.io/tutorials